(Т4)

$$\xi \sim g(x) = p\{(-1\,0)\cup(0\,1)\} + p_1(0) + p_1(2)$$

$$\int_{-\infty}^{+\infty} dF(x) = 2\int_0^1 p\,dx + 2p_1 = 1 \Rightarrow 2p + 2p_1 = 1$$

$$\square \quad \theta = p \Rightarrow p_1 = \frac{1-2\theta}{2} = 0{,}5 - \theta \Rightarrow \theta \in (0\ 0{,}5)$$

$\Rightarrow$ Вер модель

$$\xi \sim g(x\theta) = \theta\{(-1\,0)\cup(0\,1)\} + (0{,}5-\theta)\{0\} + (0{,}5-\theta)\{2\}$$

I) • $\alpha_1(\theta) = \tilde{\alpha}_1$

$$\alpha_1 = \int_{-\infty}^{\infty} x\,g(x\theta)\,dx = \cancel{\int_{-1}^{1} x\theta\,dx}^{\,0} + 0 + 2(0{,}5-\theta) = 1 - 2\theta$$

$$\tilde{\alpha}_1 = \bar{x}$$

$$\Rightarrow 1 - 2\theta = \bar{x} \Rightarrow \underline{\tilde{\theta} = 0{,}5(1-\bar{x})} \quad \text{ОММ}$$

• Несмещенность

$$M\tilde{\theta} = 0{,}5\,M(1-\bar{x}) = \frac{1}{2} - \frac{1}{2}\overbrace{M\xi}^{\alpha_1} = \cancel{\frac{1}{2}} - \cancel{\frac{1}{2}} + \theta = \theta$$

$\underline{\tilde{\theta}}$ - несмещен.

• Состоятельность

$$D\tilde{\theta} = \frac{1}{4} D(1-\bar{x}) = \frac{1}{4} D\bar{x} = \frac{1}{4n} D\xi \ (=)$$

/* $$d_2 = \int_{-1}^{1} x^2 \theta \, dx + 4(0{,}5-\theta) = \theta \frac{x^3}{3}\Big|_{-1}^{1} + 2 - 4\theta =$$

$$= 2 - \frac{10}{3}\theta$$

$$D\xi = 2 - \frac{10}{3}\theta - (1 - 4\theta + 4\theta^2) = -4\theta^2 + 1 + \frac{2}{3}\theta \quad */$$

$$(=) \ \cancel{-\frac{\theta^2}{n}}_{\to 0} + \cancel{\frac{1}{4n}}_{\to 0} + \cancel{\frac{1}{6n}}_{\to 0}\theta \xrightarrow[n\to\infty]{} 0 \Rightarrow \underline{\tilde{\theta} \text{ - состоят}}$$

• $g(x\theta) = \theta\{(-1\,0)\cup(0\,1)\} + (0{,}5-\theta)\{0\} + (0{,}5-\theta)\{2\}$

– непр. дифф. по θ

• $$\int_{-\infty}^{+\infty} \frac{\partial}{\partial\theta} g(x\theta)\,dx = \int_{-1}^{1} dx - 1 - 1 = 0$$

$$\frac{\partial}{\partial\theta}\left[\int_{-1}^{1} \theta\,dx + 2(0{,}5-\theta)\right] = \frac{\partial}{\partial\theta}\left[\cancel{2\theta} - \cancel{2\theta} + 1\right] = 0$$

– 2е усл выполн.

• $$I(\theta) = M\left[\left(\frac{\partial \ln g(x\theta)}{\partial\theta}\right)^2\right] = \int_{A(\theta)} \left(\frac{\partial \ln p(x\theta)}{\partial\theta}\right)^2 p(x\theta)\,dx +$$

$$+ \sum_k \left(\frac{\partial \ln p_k(\theta)}{\partial\theta}\right)^2 p_k(\theta)$$

$$I(\theta) = \int_{-1}^{1} \frac{1}{\theta^2}\theta\,dx + 2\frac{\cancel{0{,}5-\theta}}{(0{,}5-\theta)^{\cancel{2}}} = \frac{2}{\theta} + \frac{2}{(0{,}5-\theta)} = \frac{1}{\theta(0{,}5-\theta)}$$

– непрер на $\theta \in (0; 0{,}5)$ и >0

$\Rightarrow$ <u>Вер мод</u> <u>регулярна</u>

• Вер модель регулярн.; оценка несмещ,сн.;

$D\tilde{\theta}$ - огран. на $\forall$ компакте из Θ $\Rightarrow$ $\underline{\tilde{\theta} \text{ регул. оценка}}$

• По НКР

$$-\frac{\theta^2}{n} + \frac{1}{4n} + \frac{\theta}{6n} \geq \frac{\theta(1-2\theta)}{2n}$$

т.к. не равны, ничего сказать не можем

II) •

Ф-я правдоподобия

$$L(\theta) = \prod_{i=1}^{n} g(x_i, \theta) = \theta^{n-m}(0{,}5-\theta)^m \prod x_i$$

//] m раз встреч. {0} или {2}

$$(\ln L)' = (n-m)\frac{1}{\theta} + m\frac{(-1)}{0{,}5-\theta} = 0$$

$$\frac{1}{2}n - \frac{1}{2}m - n\theta + \cancel{m\theta} = \cancel{m\theta} \Rightarrow 2n\theta = n-m$$

$$\Rightarrow \underline{\underline{\tilde{\theta} = \frac{1}{2}\left(1-\frac{m}{n}\right)}} \quad \text{ОМП}$$

$$(\ln L)'' = \underbrace{-(n-m)\frac{1}{\theta^2}}_{<0} + \underbrace{m\frac{(-1)}{(0{,}5-\theta)^2}}_{<0} < 0 \Rightarrow \max$$

• Несмещен.

$$M\tilde{\theta} = \frac{1}{2} - \frac{1}{2}M\nu = \frac{1}{2}(1-p) = \theta \Rightarrow \text{несмещен}$$

• Состоятельность

$$D\tilde{\theta} = \frac{1}{4}D\nu = \frac{p(1-p)}{4n} = \frac{1}{\cancel{4}_2 n}(1-2\theta)\cancel{2}\theta \xrightarrow[n\to\infty]{} 0$$

$\Rightarrow$ состоят

• Вер мод регул.
оценка несмещен.
$D\tilde{\theta}$ огран. на $\forall$ компакте из Θ

$\Rightarrow$ оценка регулярна

• <u>Эффективность</u>

$$D\tilde{\theta} = \frac{\theta(1-2\theta)}{2n} = \frac{1}{n\,2}\,\theta(1-2\theta)$$

$\Rightarrow$ <u>яв-ся эффективной</u>